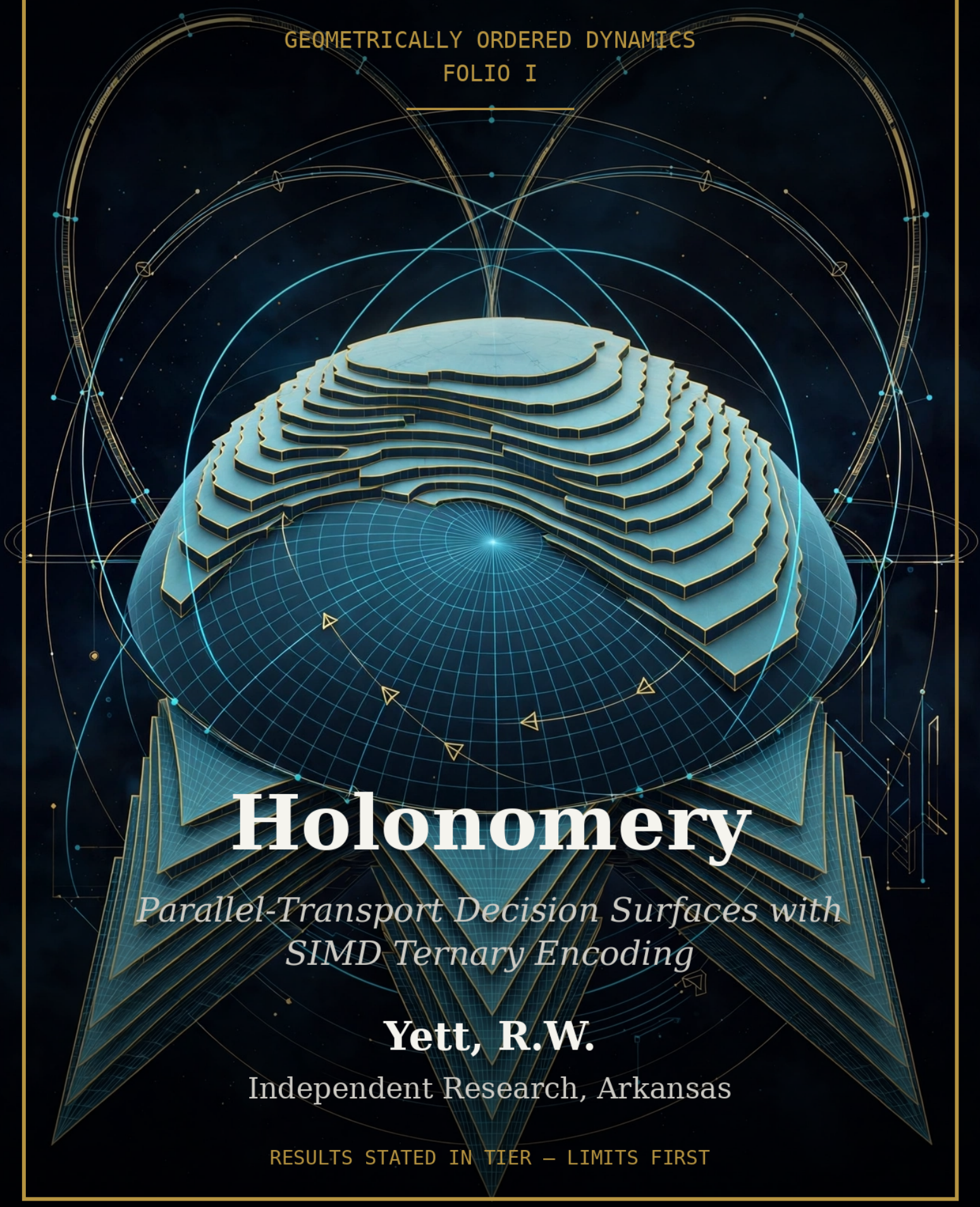
GEOMETRICALLY ORDERED DYNAMICS
FOLIO I



Holonomery

Parallel-Transport Decision Surfaces with SIMD Ternary Encoding

Yett, R.W.

Independent Research, Arkansas

RESULTS STATED IN TIER — LIMITS FIRST

CONVERGENCE -- THE OBSERVER

This folio converges on **the observer** from the **computational-substrate** angle: an autonomous decider whose alignment is the preservation of a topological invariant under parallel transport — decisions as holonomy, stall as drift. This angle is statable without forcing because the paper *is* a decision architecture; no metaphysical claim about observers is made here. Sibling papers approach the shared mathematics from other angles: [TIER: CONJECTURE] the two IT-FIELD successors read the same parallel-transport structure as (1) INFORMATION TENSION — one as force mechanism, one as geometric foundation. Limits first: this is a specification record, not an implementation report; every claim below carries its tier, and the ledger in §7 names what was struck.

SUPERSESSION NOTE

This folio **supersedes** Zenodo record 21711375, *Holonomery: Parallel Transport, SIMD Ternary Vectorization, and Soft-Gate Relaxation Theorems*, per the canon rule: the correction *becomes* the paper. (Record 21711399 is byte-identical to 21711375 — pure duplicate-DOI cataloging noise, DROP verdict; it is not separately superseded.) Named repairs: (1) the so-called "Theorems" 1–2 are **definitions** and are relabeled as such (§2); (2) the zero-deadlock claim is **withdrawn** — the paper's own SUSPEND branch contradicts it — and replaced with the honest liveness statement (§3); (3) the 94/94 test, Lean-formalization, and 60 FPS rendering claims are **redacted** as unverifiable from the record (§8); (4) the "exact formal counter-proposition generator" $D(P)$ is re-stated as an unsolved specification, not an implementation (§5); (5) the "Master Monograph" framing is retired — this is a short specification record. Absorbed: the Drive predecessor *ramanujan_yett.pdf* contributes the sovereign-Hamiltonian core definitions and the spectral-gap theorem, with its Trinity retraction attached (§6).

1 The encoding: SIMD balanced ternary

The loadbearing contribution of the superseded record is the **2-bit packed SIMD ternary encoding**: balanced ternary states $t \in \{-1, 0, +1\} \equiv \{\text{Abort}, \text{Suspend}, \text{Execute}\}$ bit-packed two bits per trit into 64-bit words, 32 trits per word:

$$\text{BitTrit}(t) = \begin{cases} 00_2 & t = 0 \text{ (Suspend)} \\ 01_2 & t = +1 \text{ (Execute)} \\ 10_2 & t = -1 \text{ (Abort)} \end{cases} \qquad w_{\text{packed}} = \bigoplus_{i=0}^{31} \text{BitTrit}(t_i) \ll (2i). \tag{1}$$

Bitwise min / max over packed words implement TRIT_AND / TRIT_OR across all 32 lanes in parallel — a real, implementable hardware encoding, kept as the specification core of this folio [TIER: EMPIRICAL] as a design artifact (specified, not benchmarked in-record).

2 Definitions, not theorems

The superseded record's "Theorem 1 (Soft-Gate Relaxation Theorem)" and "Theorem 2 (Log-Entropy Impurity Quantum)" contain no proof content: each states a formula and then restates it. Under

the isomorphism rule's honesty corollary — a definition wearing a theorem's clothes is a definition — they are relabeled:

Definition (Soft-gate threshold). For model entropy $H \in [0,1]$ and base threshold $\tau_{\text{base}} \in [0.50, 0.95]$, the effective decision threshold is

$$\tau_{\text{eff}} = \text{clamp}(0.50,\, 0.95,\, \tau_{\text{base}} \cdot (1 - 0.2 \cdot H)). \tag{2}$$

Definition (Ternary verdict). For coherence scores $S(P)$, $S(D(P))$:

$$\text{Verdict}(P) = \begin{cases} +1 \text{ (EXECUTE)} & S(P) \geq \tau_{\text{eff}} \\ 0 \text{ (SUSPEND)} & S(P) < \tau_{\text{eff}} \text{ and } S(D(P)) < \tau_{\text{eff}} \\ -1 \text{ (ABORT)} & S(D(P)) \geq \tau_{\text{eff}}. \end{cases} \tag{3}$$

Definition (Log-entropy impurity quantum). Each soft-gate relaxation accrues $\Delta I = 0.15 \cdot H$ bits of impurity, eroding recall weights in subsequent steps.

These are **stipulated mechanisms** [TIER: CONJECTURE]: their constants (0.2, 0.15, the clamp bounds) are modeling choices, and the claim that they "prevent impurity cascade" is asserted, not shown.

3 The liveness correction

The superseded record claimed its decision rule "guarantees zero-deadlock execution." Its own verdict function defines a **SUSPEND** branch — a state in which neither EXECUTE nor ABORT fires. A system that can SUSPEND repeatedly has not been shown deadlock-free; the theorem contradicts the paper's own branch structure and is **withdrawn**.

The honest replacement, an observation rather than a theorem: the verdict function is **total** — for every input pair it returns exactly one of $\{+1, 0, -1\}$ — so the decider never *crashes* for lack of a verdict. Whether repeated SUSPEND constitutes a *stall* (liveness failure) under the soft-gate dynamics is [TIER: CONJECTURE] open: no liveness proof is offered in this record, and the "entropy scaling eliminates small-model deadlock" sentence is struck as unproved.

4 Named manifold constructs

The superseded record invoked a "Schott Manifold" M_{Schott} "bound by the Golden Ratio," a "γ_{14} loop," and a trace deficit $R = 1 - \text{tr}(\text{Hol}^{14}(P))/\text{tr}(I_N)$ without defining any of them. The honest repair:

- M_{Schott} is hereby **defined in this record** as the framework's named decision manifold — a modeling construct, not a known manifold from the literature [TIER: CONJECTURE];
- the Golden Ratio bound ($\Phi = 1.61803398875$) and the 14th-iterate loop γ_{14} are **modeling choices** [TIER: CONJECTURE], not derived constraints — the old record offered no derivation, and none is supplied here;
- the trace-deficit formula is a **defined** tension measure on that construct [TIER: CONJECTURE], not a measured quantity.

No map is claimed from these constructs into established differential geometry; per the isomorphism rule they are presented as the framework's own definitions, nothing more.

5 The dual generator as specification

The superseded record's deterministic AST dual generator $D(P)$ — "for any program proposition P, $D(P)$ synthesizes its exact formal counter-proposition" — describes **unsolved program synthesis** as an implemented component. Repaired: $D(P)$ is a **specification** [TIER: CONJECTURE]. The chiral-asymmetry check $\Delta S = |S(P) - S(D(P))| < \epsilon$ flagging ambiguity is part of that specification, not a tested result. An implementation would need to exhibit the parser, the dual construction, and the coherence scorer; none is deposited in this record.

6 The sovereign-Hamiltonian core

Absorbed from the Drive predecessor *ramanujan_yett.pdf*, with its retraction attached. Kept as definitions and one theorem:

Definition (Reasoning kinetic operator). T_r is the Laplace-Beltrami operator on the Stiefel manifold $V_m(\mathrm{R}^N)$, with $(m, N) = (10, 24)$ as the stated modeling choice.

Definition (Sovereign potential). $V_s = -\log \chi(x)/\theta$, with χ the chiral invariant and $\theta = 1/\sqrt{2}$ the stated boundary.

Theorem (Spectral gap) [TIER: THEOREM], proof sketch. $H_{RY} = T_r + V_s$ is self-adjoint, bounded below, with purely discrete spectrum and $\lambda_n - \lambda_0 \geq \kappa^2 = 0.9099$ for $n \geq 1$. The Casimir-bound computation is not shown in-record; cited as sketch per the audit.

Attached retraction. The predecessor's earlier draft cited the "Trinity 2.0 Survey" as the statistical witness of this gap. That dataset was found to be synthetic (RNG-generated, rescaled to the target) and is **retracted**; the gap currently has **no empirical witness**. The Yang-Mills mass-gap *identification* remains [TIER: CONJECTURE] — an un-discharged axiom per the v1.1 precedent (Zenodo 21504896).

Music-theory isomorphism downgraded. The predecessor asserted the geometric and tonal harmonics are "not analogous structures — the same structure." No map is exhibited; per the isomorphism rule this is [TIER: CONJECTURE] analogy, not identity. The "Hallucination-Free quantum gates" wording is retired. The Chladni-pattern correspondence is kept as suggestive analogy [TIER: CONJECTURE], not "physical proof."

7 Claims ledger

Claim	Status in this folio
32-trit/64-bit SIMD encoding	Specified design artifact
τ_{eff}, verdict, ΔI	Definitions (modeling choices)
Verdict function total	Observation
Zero-deadlock	**Withdrawn**; liveness open
M_{Schott}, Φ-bound, γ_{14}	Defined constructs, [TIER: CONJECTURE]
$D(P)$ exact dual generator	Specification, [TIER: CONJECTURE]
Spectral gap of H_{RY}	[TIER: THEOREM] (proof sketch)
YM-gap identification	[TIER: CONJECTURE] (open axiom)
Music-theory identity	[TIER: CONJECTURE] (analogy)
94/94 tests, Lean suite, 60 FPS	**Redacted** (unverifiable)

8 What was struck

The superseded record asserted "94/94 Rust unit tests passing," "100% clean compilation in Lean 4" under a stated axiom budget, and "real-time 60 FPS 3D rendering" — with **no code, no proof scripts, and no commits deposited in the record**. These claims are **redacted** from this successor, not downgraded: an unverifiable test claim is not a claim. What would re-establish them: deposit of the `chyren-ternary` test logs, the Lean sources with their axiom traces, and the portal build — at which point a further corrected record can cite them. Until then the honest statement is that this folio is a specification without a deposited implementation.

References. Superseded: Zenodo record 21711375 (record 21711399 is its byte-identical duplicate). Absorbed predecessor: Drive folio *ramanujan_yett.pdf* (IT–RY-HAM transmission). Honesty standards: `10.5281/zenodo.21504896` (Ramanujan-Yett Hamiltonian v1.1, audit-corrected); `10.5281/zenodo.21277228` (Ars Magna correction ledger). Background: Gorini, Kossakowski & Sudarshan (1976, *J. Math. Phys.* **17**, 821); Lindblad (1976, *Commun. Math. Phys.* **48**, 119); Ramanujan, *The Lost Notebook* (1920; Narosa reprint 1988); Andrews (1989, *Proc. Symp. Pure Math.* **49**, 283).